

## GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>Project</b>                 | FEED for Gas Gathering Pipelines & Associated Infrastructure |
| <b>Client</b>                  | Brass Fertilizer and Petrochemical Company Limited           |
| <b>Originating Company</b>     | Epic Atlantic Limited                                        |
| <b>Document Title</b>          | Geotechnical Survey Review Report                            |
| <b>Document Number</b>         | EA-BFPCL-GPMC-GGPL-FEED-CIV-RPT-002                          |
| <b>Document Revision</b>       | R01                                                          |
| <b>Document Status</b>         | Issued For Information                                       |
| <b>Originator</b>              | Onwuka Idika                                                 |
| <b>Security Classification</b> | Restricted                                                   |
| <b>Issue Date</b>              | 09 May 2022                                                  |

## Geotechnical Survey Review Report

|       |            |                        |              |               |                 |                  |
|-------|------------|------------------------|--------------|---------------|-----------------|------------------|
|       |            |                        |              |               |                 |                  |
|       |            |                        |              |               |                 |                  |
|       |            |                        |              |               |                 |                  |
|       |            |                        |              |               |                 |                  |
|       |            |                        |              |               |                 |                  |
| R01   | 09-05-22   | Issued for Information | Onwuka Idika | Williams Irem | Kazeem Yusuff   |                  |
| Rev # | Issue Date | Status Description     | Originator   | Reviewer      | Approver (Epic) | Approver (BFPCL) |

## Revision History

| Rev. No. | Date     | Description of Revision |
|----------|----------|-------------------------|
| R01      | 09-05-22 | Issue for Information   |
|          |          |                         |
|          |          |                         |
|          |          |                         |
|          |          |                         |

## Review Code &amp; Status (to be filled by Client)

|                                        |                                                                                            |              |
|----------------------------------------|--------------------------------------------------------------------------------------------|--------------|
| <input type="checkbox"/> <b>Code 1</b> | <b>Approved</b> – Further work may proceed.                                                |              |
| <input type="checkbox"/> <b>Code 2</b> | <b>Approved with comments.</b> – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed |              |
| <input type="checkbox"/> <b>Code 3</b> | <b>Reviewed</b> – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed            |              |
| <input type="checkbox"/> <b>Code 4</b> | <b>Retained</b> for information only                                                       |              |
| <input type="checkbox"/> <b>Code 5</b> | <b>Review Not Required</b> – Do not Resubmit                                               |              |
| <b>Date:</b>                           | <b>Position:</b>                                                                           | <b>Sign:</b> |

## Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)  
*for the subsequent project phases, post FEED, it is expected that:*
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

## Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

### SUMMARY PAGE

|                  |                                                     |                                                                  |
|------------------|-----------------------------------------------------|------------------------------------------------------------------|
| <b>Client:</b>   | Brass Fertilizer & Petrochemical Company Limited    |                                                                  |
| <b>Project:</b>  | Gas Processing & Methanol Complex                   | <b>Project No.</b> 11998A                                        |
| <b>Facility:</b> | Gas Gathering Pipelines & Associated Infrastructure | <b>Doc No.</b><br>EA-BFPCL-GPMC-GGPL-FEED-CIV-RPT-002CIV-RPT-002 |

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

**KEYWORDS:** BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

| REFERENCES   |                                                                |
|--------------|----------------------------------------------------------------|
| Document No. | Description                                                    |
|              | Draft Final Report - Right of Way Geotechnical Survey by BRONE |
|              |                                                                |
|              |                                                                |
|              |                                                                |
|              |                                                                |
|              |                                                                |
|              |                                                                |

## Table of Contents

|                                                            |    |
|------------------------------------------------------------|----|
| PROJECT INFORMATION, TERMS, ABBREVIATIONS, AND DEFINITIONS | 6  |
| Project Information – General                              | 6  |
| Project Information – General (Contd)                      | 7  |
| Definition of Terms                                        | 8  |
| Abbreviations                                              | 8  |
| 1. INTRODUCTION                                            | 9  |
| 1.1. Background Information                                | 9  |
| 1.2. Purpose of Document                                   | 11 |
| 1.2.1. The Phase 1 Facilities                              | 11 |
| 1.2.2. The Phase 2 Facilities                              | 12 |
| 1.3. The Phase 1 Pipelines Route / Right of Way            | 12 |
| 1.4. Scope                                                 | 12 |
| 1.5. Design Battery limits                                 | 13 |
| 1.5.1. BBOU Platform                                       | 13 |
| 1.5.2. ELEP Platform                                       | 13 |
| 1.5.3. Akabel Facility                                     | 13 |
| 2. REGULATIONS, CODES AND STANDARDS                        | 14 |
| 2.1. General                                               | 14 |
| 2.2. Regulations                                           | 14 |
| 2.2.1. Local Regulations                                   | 14 |
| 2.3. Concept Engineering Referenced Documents              | 14 |
| 2.4. Shell Design and Engineering Practices (DEPs)         | 14 |
| 2.5. International Codes And Standard                      | 15 |
| 2.6. Conflict of Information                               | 15 |
| 3. GENERAL DESIGN DATA                                     | 16 |
| 3.1. Design Units of Measurement                           | 16 |
| 3.2. Standard Process Conditions                           | 16 |
| 3.3. Site and Environmental Conditions                     | 16 |
| 3.3.1. Air Temperature                                     | 16 |
| 3.3.2. Weather                                             | 16 |
| 3.3.3. Rainfall                                            | 17 |
| 3.3.4. Relative Humidity                                   | 17 |
| 3.3.5. Wind Speed and Direction                            | 17 |
| 3.3.6. Seismic Zone                                        | 17 |
| 3.4. Design Life                                           | 17 |
| 4. COMMENTS/FINDINGS                                       | 18 |
| 5. CONCLUSION                                              | 20 |

## Project Information, Terms, Abbreviations, and Definitions

### Project Information – General

|     |                                    |                                                                                                                                           |
|-----|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0 | Employer                           | BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)                                                                                  |
| 2.0 | PMC                                | TATA CONSULTING ENGINEERS LIMITED (TCE)                                                                                                   |
| 3.0 | EPCC (CONTRACTOR)                  | CHINA TIANCHEN ENGINEERING CORPORATION (TCC)                                                                                              |
| 4.0 | Project Title                      | Gas Processing and Methanol Complex                                                                                                       |
| 5.0 | Location                           | Akabel, Brass Island.<br>The project site is about 85km South West of Port Harcourt, 70km South of Yenagoa and 90km West of Bonny Island. |
| 5.1 | State                              | Bayelsa State, Nigeria                                                                                                                    |
| 5.2 | Nearest Air Port and distance      | Port Harcourt Airport (PHC)<br>Bayelsa Airport (Under Construction)                                                                       |
| 5.3 | Nearest Port                       | Port Harcourt (Onne)                                                                                                                      |
| 5.4 | Nearest Railway Station & distance | -                                                                                                                                         |
| 5.5 | Highway and distance               | Connected through water way only. Road connection does not exist.                                                                         |
| 5.6 | Important Town and distance        | Yenagoa, Nigeria & 67 km                                                                                                                  |
| 5.7 | Altitude Above Mean Sea Level      | 9 m                                                                                                                                       |
| 5.8 | Atmospheric pressure               |                                                                                                                                           |
|     | Average                            | 1013 mbar                                                                                                                                 |
| 6.0 | Ambient Air Temperatures           |                                                                                                                                           |
| 6.1 | Maximum                            | 38°C                                                                                                                                      |
| 6.2 | Minimum                            | 18°C                                                                                                                                      |
| 6.3 | Average                            | 23°C                                                                                                                                      |
| 6.4 | Summer Design Wet Bulb Temp        | 28°C                                                                                                                                      |
| 6.5 | Winter Design Wet Bulb Temp        | 15°C                                                                                                                                      |
| 6.6 | MDMT                               | 15°C                                                                                                                                      |

### Project Information – General (Contd)

|      |                                                     |                                                                   |
|------|-----------------------------------------------------|-------------------------------------------------------------------|
| 6.7  | Swamp temperature                                   | 25°C. (23°C. to be used wherever a margin is required)            |
| 6.8  | Ground temperature                                  | 25 - 27.5°C.                                                      |
| 6.9  | Black-bulb temperature:                             | 80°C.                                                             |
| 7.0  | Relative Humidity                                   |                                                                   |
| 7.1  | Maximum                                             | 95 %                                                              |
| 7.2  | Minimum                                             | 51 %                                                              |
| 7.3  | Average                                             | 73 %                                                              |
| 8.0  | Rainfall Data                                       |                                                                   |
| 8.1  | Average annual rainfall                             | 3800mm                                                            |
| 8.2  | Max daily rainfall:                                 | 180mm                                                             |
| 8.3  | Mean max. Hourly rainfall                           | 100mm                                                             |
| 9.0  | Transport                                           |                                                                   |
| 9.1  | Name of the highway near which the plant is located | Connected through water way only. Road connection does not exist. |
| 9.2  | Access road                                         | -                                                                 |
| 9.3  | Railway (Gauge)                                     | -                                                                 |
| 10.0 | Wind Data                                           |                                                                   |
| 10.1 | Design Wind Velocity                                | 35.6 m/sec (Note-1)                                               |
| 10.2 | Wind Direction                                      |                                                                   |
| 11.0 | Seismic Data                                        |                                                                   |
| 11.1 | Earthquake Zone                                     |                                                                   |
| 12.0 | High Flood Level (HFL)                              | *                                                                 |
| 13.0 | Natural Ground Level                                | *                                                                 |
| 14.0 | Slope                                               | *                                                                 |
| 15.0 | Drainage                                            | *                                                                 |
| 16.0 | Design temperature for Electrical equipment         | 40°C                                                              |

## Definition of Terms

| TERMS                 | DEFINITIONS                                                                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BFPCL                 | Brass Fertilizer & Petrochemical Limited.                                                                                                                                                                                                                                        |
| SPDC                  | Shell Petroleum Development Company Nig. Limited                                                                                                                                                                                                                                 |
| Employer              | Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer. |
| Engineer              | The party that carries out the FEED Engineering of the project.                                                                                                                                                                                                                  |
| Manufacturer/Supplier | Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.                                                                                                                                                                  |
| Shall                 | The word 'shall' indicate a mandatory requirement.                                                                                                                                                                                                                               |
| Should                | The word 'should' indicate a recommendation.                                                                                                                                                                                                                                     |
| Measuring Range       | Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).                                                                                                                                |
| On-Line Instruments   | Instruments connected to process and utility lines or equipment via primary isolation valves.                                                                                                                                                                                    |
| Project               | Front End Engineering Design for Gas Gathering Pipelines (Phase1)                                                                                                                                                                                                                |

## Abbreviations

| ABBREVIATION | MEANING |
|--------------|---------|
|              |         |
|              |         |
|              |         |



# 1. Introduction

---

## 1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a greenfield methanol production facility in Akabel (Odioma), Brass LGA Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields, namely: Bubouwe Bou & Elepa, in OML 33, Nun River, Seibou, in OML 35, and Opugbene in OML 36.

The Project involves the development, construction, and operation of a 10,000 metric tonnes per day (MTPD) capacity methanol production plant comprising the following elements:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel, - as shown in Figures 1.1, 1.2, & 1.3.
- Product Export Lines and associated Offshore Single Buoy Mooring (SBM) for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) which are required for the operation of the processing plants.

Tata Consulting Engineers Limited (TCE) is the Project Management Consultant for the project.

The Nodal Gas Gathering Schematics

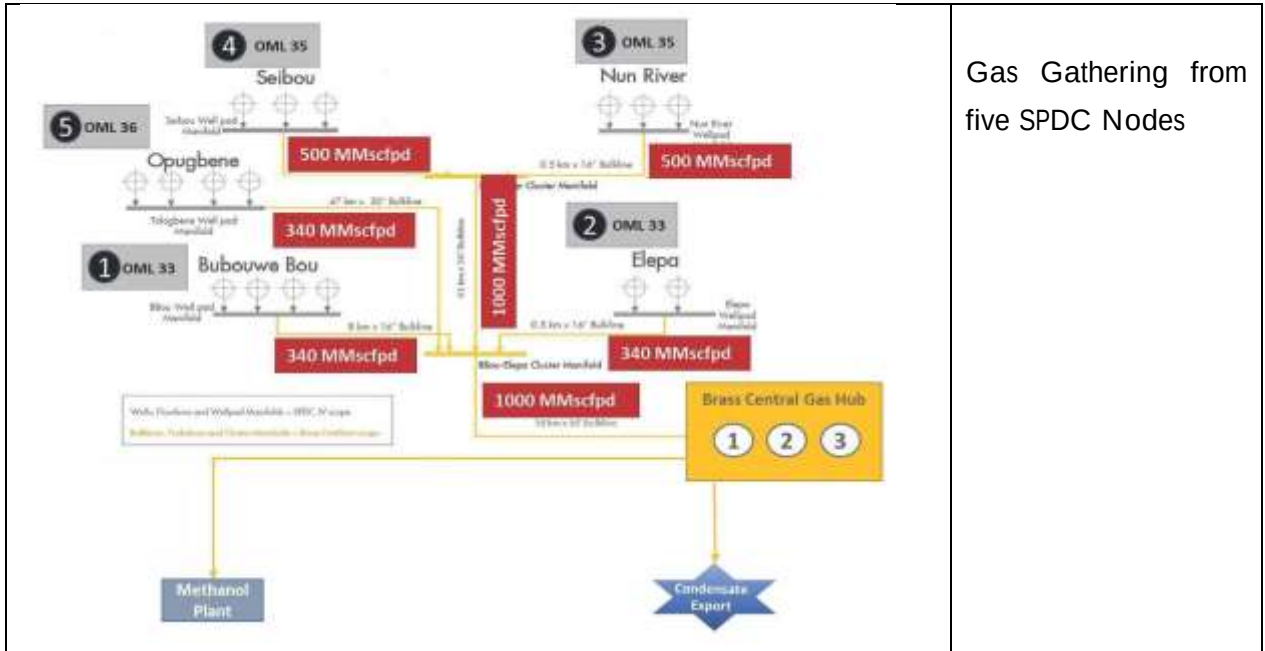
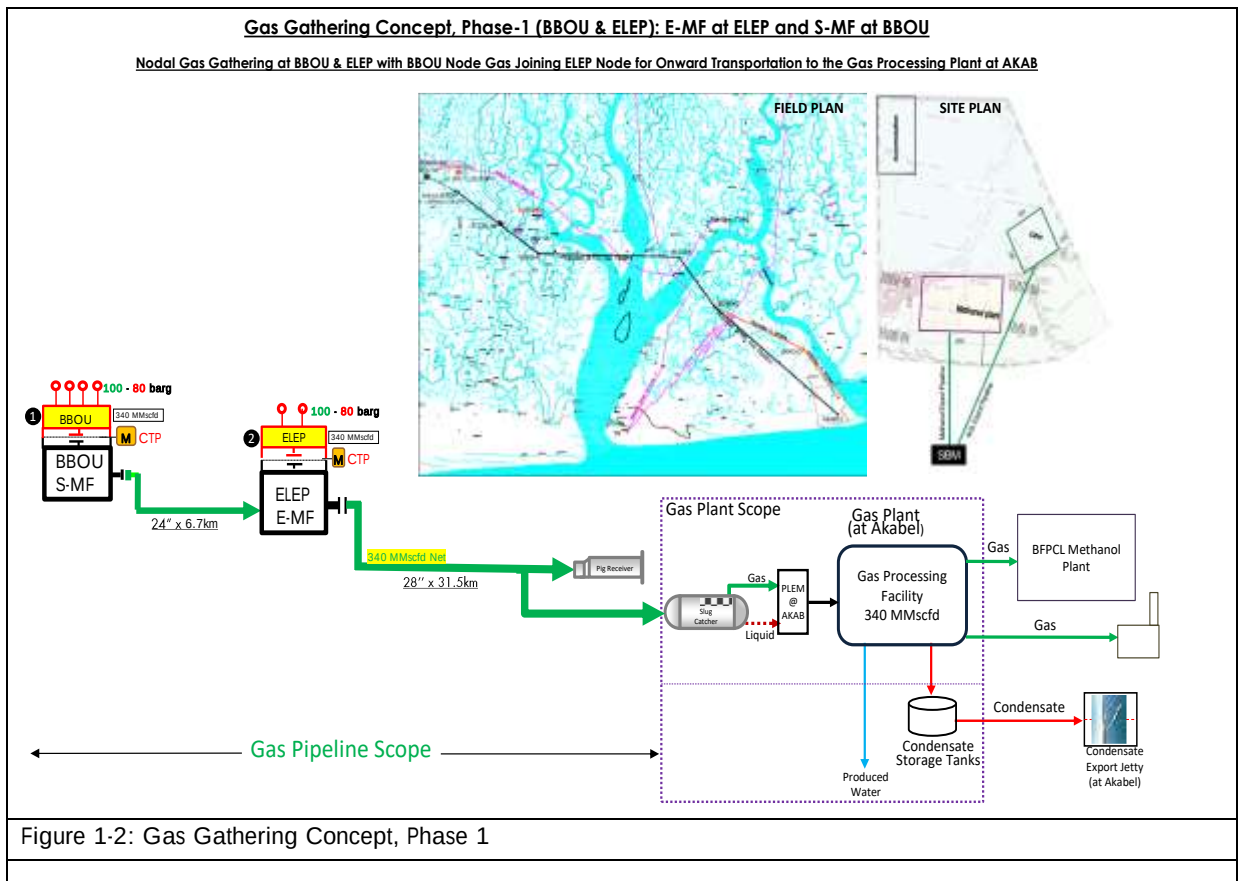


Figure 1-1: Field &amp; Nodes covered in the Gas Gathering

## The Nodal Gas Gathering Schematics (contd.)



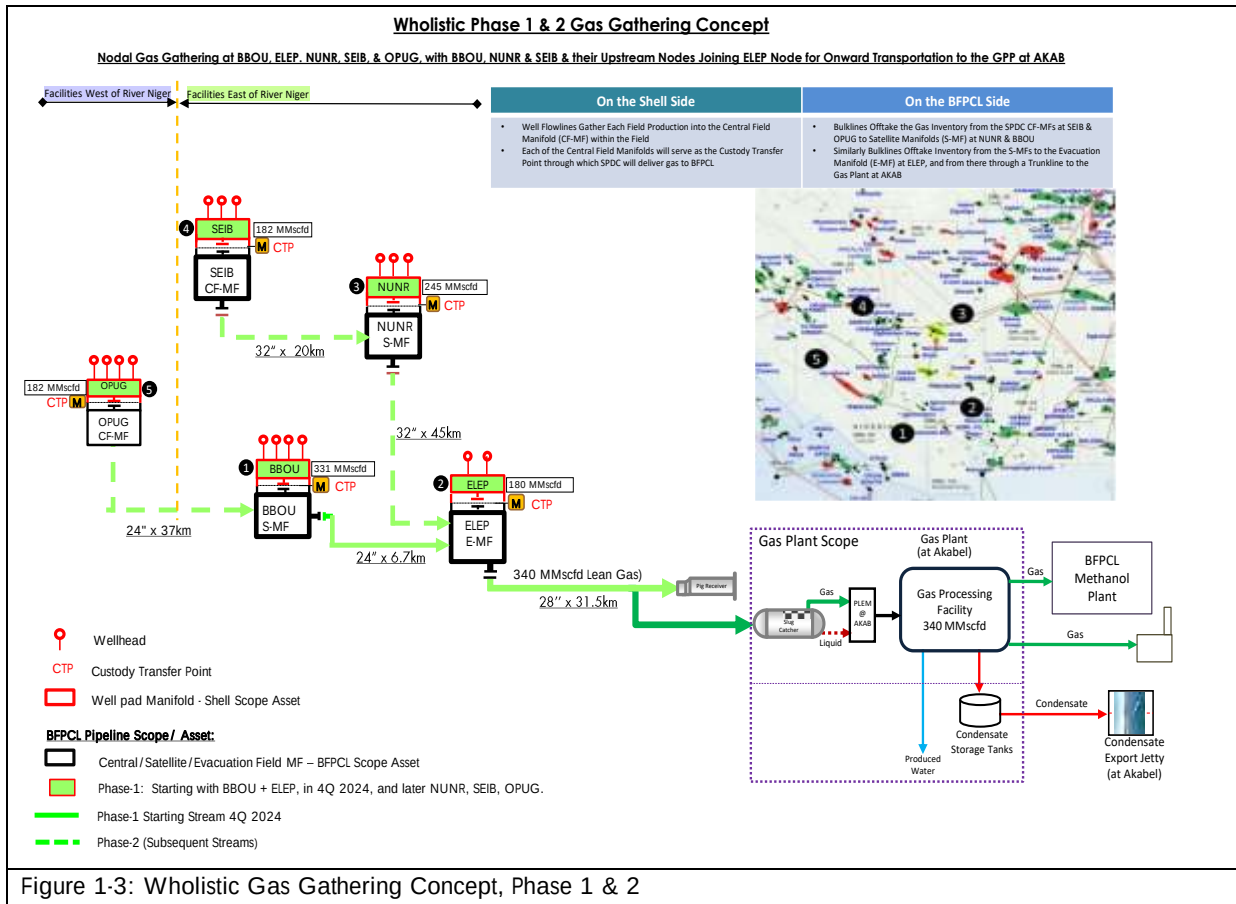


Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 &amp; 2

## 1.2. Purpose of Document

The purpose of this document is to provide feedback to TCE/BFPCL on the adequacy of the geotechnical survey carried out by Messrs BRONE.

The design and construction of the pipelines shall fully comply with all relevant contractual requirements specified in the Scope of Work and Client's Standards.

The gas gathering will be in two phases, as shown in the schematics below:

- Phase 1 comprising gas gathering from Bubouwe Bou and Elepa fields, and
- Phase 2 comprising gathering from Nun River, Seibou, and Opugbene fields.

### 1.2.1. The Phase 1 Facilities

The phase 1 facilities (shown in Figure 1.1) are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU satellite Manifold
- ELEP evacuation Manifold, and
- The Pipeline End Manifold at AKAB.

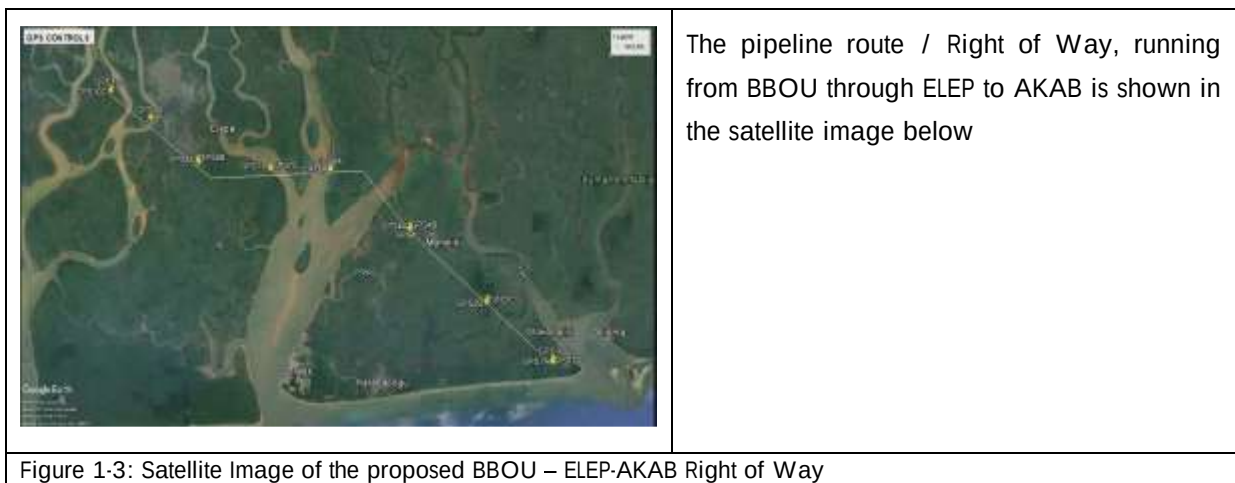
These phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED. The phase 2 pipelines routes are yet to be defined and surveyed.

#### 1.2.2. The Phase 2 Facilities

The phase 2 facilities comprise NUNR, SEIB, and OPUG Manifolds with their pipelines connecting to phase 1 facilities as shown in Figure 1.2.

#### 1.3. The Phase 1 Pipelines Route / Right of Way



#### 1.4. Scope

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEM complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEM
- Pipeline from ELEM to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEM Manifold
- PIG Receivers at ELEM and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.

- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

The Scope of the FEED Engineering covers the provision of all necessary information, technical drawings, philosophies, specifications, design data, listing of codes and standards required for the detailed design and construction of the Multiphase pipeline system.

## **1.5. Design Battery limits**

The battery limits for this Phase 1 FEED Project are:

### **1.5.1. BBOU Platform**

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEM.

### **1.5.2. ELEM Platform**

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

### **1.5.3. Akabel Facility**

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

## 2. Regulations, Codes And Standards

---

### 2.1. General

The regulations, codes, and standards to be applied shall always be the latest revision thereof and any amendment or supplement thereto in effect as at the contract date with contractor, manufacturer, or supplier. Design, fabrication, and installation work will be governed by the following regulations, codes, and standards.

### 2.2. Regulations

Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. Guidance on the statutory requirements for pipelines design, construction and operation should be obtained from the following documents:

#### 2.2.1. Local Regulations

| Regulation | Function |
|------------|----------|
|            |          |

### 2.3. Concept Engineering Referenced Documents

| Document Number | Title |
|-----------------|-------|
|                 |       |
|                 |       |

### 2.4. Shell Design and Engineering Practices (DEPs)

Applicable DEPs, supplements and MESC SPE Codes are as referenced in various discipline dossiers. The design shall be based on DEP Feb 2019 (or latest, as advised by the EMPLOYER) some of which are listed below.

| Document Number | Document Title |
|-----------------|----------------|
|                 |                |

## 2.5. International Codes And Standard

| DISCIPLINE | DOCUMENT NUMBER | TITLE |
|------------|-----------------|-------|
|            |                 |       |
|            |                 |       |

## 2.6. Conflict of Information

The edition of each document current at the date of enquiry shall apply, together with all published amendments. Where conflict occurs between the requirements of this Specification and referenced Codes and Standards, the Engineer shall notify EMPLOYER in writing immediately for resolution. In all cases, the most stringent shall apply. All deviations from the requirements of these Specifications, and the appendices and the referenced Codes and Standards shall be stated on a Technical Query (TQ) and issued to Employer for approval. In the absence of such a TQ, full compliance with the order of precedence below shall be assumed. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

### 3. General Design Data

---

#### 3.1. Design Units of Measurement

| Variable | Unit |
|----------|------|
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |
|          |      |

#### 3.2. Standard Process Conditions

Standard process conditions shall be taken as 15°C and 1.01325 bar.

#### 3.3. Site and Environmental Conditions

The following site and environmental conditions have been extracted from Referencing paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited.

##### 3.3.1. Air Temperature

Average maximum (daytime) temperature in the dry season months was about 32°C, and about 28°C in the rainy season. The minimum (night-time) temperature level was fairly the same (23°C) through-out the year.

##### 3.3.2. Weather

Two weather conditions are prevalent in the region:

Dry Season (Harmattan)



During Harmattan season (November – January, 2-3 months/year), the atmosphere is laden with extremely fine dust (particle size of 5 microns or less), which can penetrate equipment, block air filters, etc.

#### Wet Season

Heavy rainfall can be expected in the wet season which normally runs from the months of April through October. Lightning is very frequent especially during the rainy season and shall be specifically considered in the design of all relevant equipment.

#### 3.3.3. Rainfall

Rainfall steadily increases from about 65mm in January to about 288mm in July after which it drops to about 245mm in August. This amount increases to a second maximum peak of about 331mm in September and then steadily drops to 47mm in December. The double maxima characteristics of Equatorial climate are very evident in the project area.

#### 3.3.4. Relative Humidity

| Parameters | Values |
|------------|--------|
| Mean       | 81%    |
| Minimum    | 69%    |
| Maximum    | 88%    |

#### 3.3.5. Wind Speed and Direction

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

The south-westerly wind is predominant (about 60% of the annual winds) and prevails during the wet season (March - October). The north-easterly wind which predominate during the dry season (November– February) and also brings the harmattan accounts to about 32% of the annual winds.

#### 3.3.6. Seismic Zone

The project area is classified as seismic zone 0.

### 3.4. Design Life

All equipment for the project shall be designed based on a service life of 30 years.

## 4. COMMENTS/FINDINGS

The geotechnical survey as carried out by M/s BRONE have investigation boreholes numbered BH1 to BH 75, with a commencement point from the methanol plant location (coordinate of BH1, Easting of 211214.60 (m) and Northing of 477124.60 (m)) to the former location of the Bou-bouwei M/F (coordinate of BH75, Easting of BH 75 181356.62 (m) and Northing of 493133.06 (m) covering a distance of 35.68 Km. See page 12 of report titled " Draft Final Report"

Having reviewed holistically the above stated Geotechnical report, EPIC ATLANTIC hereby state the following under listed as our final position on the document.

| Sr N | Description                                                                                                                                                                          | Report Section       | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Pipeline End Manifold locations are to be installed in three locations of the RoW at Akabel, Elepa and Bou-bouwei Bou                                                                | General              | <ul style="list-style-type: none"> <li>•No specific soil investigation carried out for the three manifold slot locations.</li> </ul>                                                                                                                                                                                                                                                                                                                  |
| 2    | Soil test laboratory                                                                                                                                                                 | General              | <ul style="list-style-type: none"> <li>•BRONE nominated KYJE to carry out the soil testing, however there is no indication of the performance of the soil laboratory in the whole works.</li> <li>•There is no laboratory custody transfer of the samples.</li> </ul>                                                                                                                                                                                 |
| 3    | BH 1 location (start point of soil investigation) is assumed in the Geotechnical report to be at the Methanol plant location                                                         | Page 290 to Page 292 | <ul style="list-style-type: none"> <li>•No exact location soil investigation carried out at the position for the Akabel manifold area.</li> <li>•Pipeline end manifold required to be within the area GPP location</li> </ul>                                                                                                                                                                                                                         |
| 4    | The former (Before SPDC indicated a new location for the Boubouwei Bou manifold) BH 75 location (end point of soil investigation) is assumed to be within the BouBouwei Bou manifold |                      | <ul style="list-style-type: none"> <li>•No soil profile information at areas within the "former BouBouwei Bou manifold" that can be used for foundation design. See area tagged "D" in page 289 of report</li> <li>•Pipeline end manifold is required to be a slot off the Pipeline RoW</li> <li>•SPDC have given a new location which is 2.3km farther north west of RoW and no soil investigation information is available for the area.</li> </ul> |
| 5    | The location for the ELEPA manifold as assumed in the Geotechnical is to be around BH 55 area                                                                                        | Page 11 and 289      | <ul style="list-style-type: none"> <li>•No soil investigation parameters within the ELEPA manifold slot area that can be used for foundation design. See area tagged "C" in page 289 of report.</li> </ul>                                                                                                                                                                                                                                            |

| Sr N | Description                                      | Report Section       | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | Pile foundation design for the manifold platform | Page 168 to page 180 | <ul style="list-style-type: none"> <li>•Lack of pile design parameter using SPT. SPT values range from freefall (0) to maximum of 2-count (as recorded)</li> <li>•No design strength parameters such as angle of internal friction in degrees for the slots of the manifold &amp; undrained cohesion or shear strength per m<sup>2</sup> from top to depth of termination of investigation</li> <li>•No calculation of settlement of the pile either for a single pile or group founded on such soft clay strata</li> </ul>              |
| 7    | Bearing capacity                                 | Page 48              | <ul style="list-style-type: none"> <li>•The parameters of the soil that were investigated in the report are not used to compute the pile bearing capacity</li> <li>•The bearing capacity as depicted in the report does not give a true reflection of the property of soils investigated</li> <li>•The written bearing capacity are typical generally for soil formations of dense sand and not of peaty soft clay</li> </ul>                                                                                                            |
| 8    | Design for shallow foundations                   | Page 45              | <ul style="list-style-type: none"> <li>•The depth of 1 to 3 m bearing is not reliable because sample is so soft that it cannot be collected on a UD tube or remoulded to be performed on undrained tri-axial test for bearing capacity</li> <li>•The bearing capacity consideration for shallow foundation does not depict soil lithology at that depth, there is almost flow-able soft mud at the upper layer and hence cannot perform undrained tri-axial test.</li> </ul>                                                             |
| 9    | Sand formation at BH 1,2,9                       | Page 290             | <ul style="list-style-type: none"> <li>•No Geological model to show lateral and vertical extent of soil strata on the investigated locations, including the 'sand' formation as depicted in BH 1,2,9 profile, hence the "sand" formation couldn't have just been profiled to terminate exactly at every 500 meters as depicted in the lithology</li> <li>•The underlying bearing strata immediately below the "sand" is peaty soft clay and the idea of using peaty soft clay to bear loads is disastrous and unprofessional.</li> </ul> |

## 5. CONCLUSION

---

Based on the available geotechnical information, it cannot be technically justified to use the soil bearing capacities that is copied and written down in the final report as a geotechnical parameter of design for the foundations for the

1. Manifolds at the three locations as there is no soil bearing capacity for the mentioned three manifold locations.
2. It is imperative that Manifold foundation design will be fully carried out at detailed design stage after the pre-engineering survey is carried out.
3. Meanwhile, industry-based criteria will be used to establish the foundation costs as input to FEED deliverables.